CHAPTER 9

Organic Chemistry- Some Basic Principles and Techniques

Tetravalence of Carbon (Shapes of organic compounds)

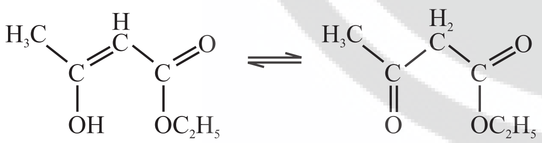
1. Which of the following molecules represents the order of hybridisation sp^2, sp^2, sp, sp from left to right atoms? *(2018)*

a. $HC \equiv C - C \equiv CH$
b. $CH_2 = CH - C \equiv CH$
c. $CH_3 - CH = CH - CH_3$
d. $CH_2 = CH - CH = CH_2$

2. Which of the following species contains equal number of σ-bonds and π-bonds? *(2015)*

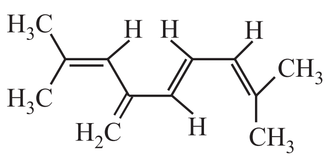
a. $(CN)_2$
b. $CH_2(CN)_2$
c. HCO_3^-
d. XeO_4

3. The enolic form of ethyl acetoacetate as shown below has *(2015)*



a. 9 σ-bonds and 2 π-bonds
b. 9 σ-bonds and 1 π-bonds
c. 18 σ-bonds and 2 π-bonds
d. 16 σ-bonds and 1 π-bond

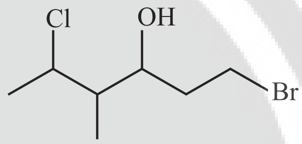
4. The total number of π-bond electrons in the following structure is *(2015)*



a. 12
b. 16
c. 4
d. 8

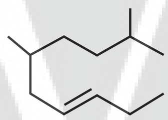
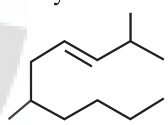
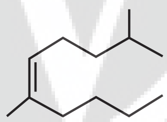
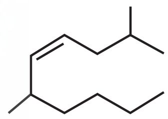
Classification, Nomenclature of Organic Compounds

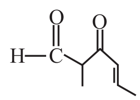
5. The correct IUPAC name of the following compound is: *(2022)*



a. 6-bromo-4-methyl-2-chlorohexan-4-ol
b. 1-bromo-5-chloro-4-methylhexan-3-ol
c. 6-bromo-2-chloro-4-methylhexan-4-ol
d. 1-bromo-4-methyl-5-chlorohexan-3-ol

6. The correct structure of 2,6-Dimethyl-dec-4-ene is: *(2021)*

a. 
b. 
c. 
d. 

7. The IUPAC name of the compound 

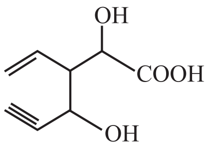
is *(2017-Delhi)*

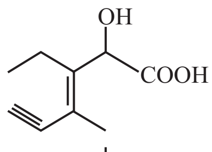
a. 3–keto–2–methylhex–5–enal
b. 3–keto–2–methylhex–4–enal
c. 5–formylhex–2–en–3–one
d. 5–methyl–4–oxohex–2–en–5–al

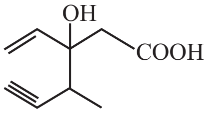
8. The structure of isobutyl group in an organic compound is: *(2013)*

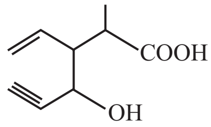
a. $(CH_3)_2 > CH-CH_2-$
b. $CH_3-\underset{|}{C}H-CH_2-CH_3$
c. $CH_3-CH_2-CH_2-CH_2-$
d. $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-$

9. Structure of the compound whose IUPAC name is 3-Ethyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid is: *(2013)*

a. 

b. 

c. 

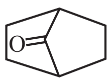
d. 

Isomerism

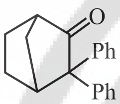
10. The compound which shows metamerism is: *(2021)*

a. C_3H_8O b. C_3H_6O
c. $C_4H_{10}O$ d. C_5H_{12}

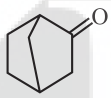
11. Which among the given molecules can exhibit tautomerism? *(2016 - II)*



I



II



III

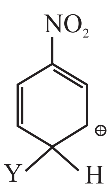
a. III only b. Both I and III
c. Both I and II d. Both II and III

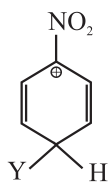
Fundamental Concepts of Organic Reactions

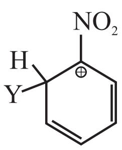
12. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following? *(2020)*

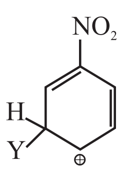
a. + R effect of – CH_3 groups
b. – R effect of – CH_3 groups
c. Hyperconjugation
d. –I effect of – CH_3 groups

13. Which of the following carbocations is expected to be most stable? *(2018)*

a. 

b. 

c. 

d. 

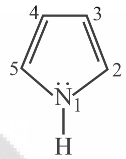
14. Which of the following is correct with respect to – I effect of the substituents? (R = alkyl) *(2018)*

a. – NH_2 < – OR < – F b. – NR_2 < – OR < – F
c. – NR_2 > – OR > – F d. – NH_2 > – OR > – F

15. The correct statement regarding electrophile is: *(2017-Delhi)*

a. Electrophile can be either neutral or positively charged species and can form a bond by accepting a pair of electrons from a nucleophile
b. Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from a nucleophile
c. Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from another electrophile
d. Electrophiles are generally neutral species and can form a bond by accepting a pair of electrons from a nucleophile

16. In pyrrole, the electron density is maximum on *(2016 - II)*

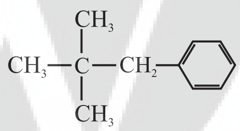


a. 2 and 3 b. 3 and 4
c. 2 and 4 d. 2 and 5

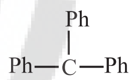
17. Which of the following statements is not correct for a nucleophile? *(2015 Re)*

a. Nucleophiles are not electron seeking
b. Nucleophile is a Lewis acid
c. Ammonia is a nucleophile
d. Nucleophiles attack low electron density sites

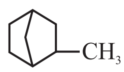
18. Consider the following compounds



(I)



(II)

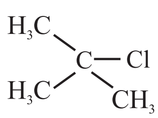


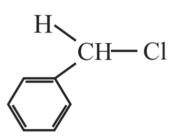
(III)

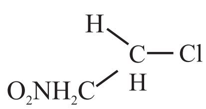
Hyperconjugation occurs in: *(2015)*

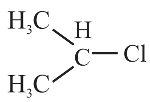
a. II only b. III only
c. I and III d. I only

19. In which of the following compounds, the C – Cl bond ionization shall give most stable carbonium ion? *(2015)*

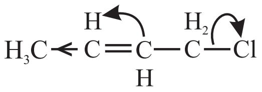
a. 

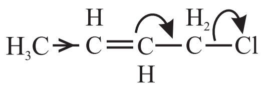
b. 

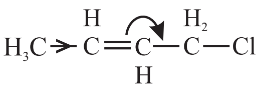
c. 

d. 

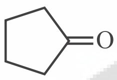
20. Which of the following is the most correct electron displacement for a nucleophilic reaction to take place? *(2015)*

a. 

b. 

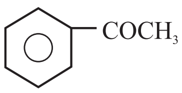
c. 

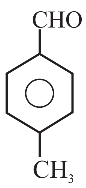
d. 

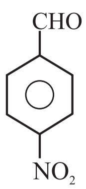
21. Treatment of cyclopentanone  with methyl lithium gives which of the following species? *(2015)*

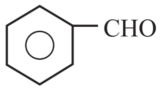
a. Cyclopentanonyl radical
b. Cyclopentanonyl biradical
c. Cyclopentanonyl anion
d. Cyclopentanonyl cation

22. Which one is most reactive towards nucleophilic addition reaction? *(2014)*

a. 

b. 

c. 

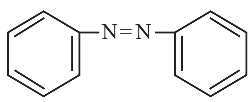
d. 

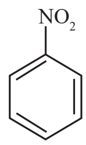
23. Some *meta*-directing substituents in aromatic substitution are given. Which one is most deactivating? *(2013)*

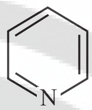
a. — COOH
b. — NO_2
c. — $C \equiv N$
d. — SO_3H

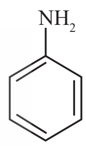
Methods of Purification, Qualitative and Quantitative Analysis of Organic Compounds

24. The Kjeldahl's method for the estimation of nitrogen **can be used** to estimate the amount of nitrogen in which one of the following compounds? *(2022)*

a. 

b. 

c. 

d. 

25. Paper chromatography is an example of : *(2020)*

a. Partition chromatography
b. Thin layer chromatography
c. Column chromatography
d. Adsorption chromatography

26. A liquid compound (x) can be purified by steam distillation only if it is *(2020-Covid)*

a. Not steam volatile, miscible with water
b. Steam volatile, miscible with water
c. Not steam volatile, immiscible with water
d. Steam volatile, immiscible with water

27. The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is *(2017-Delhi)*

a. Steam distillation
b. Sublimation
c. Chromatography
d. Crystallisation

28. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is: *(2015)*

a. 18.20
b. 16.76
c. 15.76
d. 17.36

29. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralised 10 mL of 1 M H_2SO_4. The percentage of nitrogen in the soil is *(2014)*

a. 45.33
b. 35.33
c. 43.33
d. 37.33

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	d	c	d	b	d	b	a	b	c	a	c	d	a,b	a	d	b
18	19	20	21	22	23	24	25	26	27	28	29					
b	a	b	c	c	b	d	a	d	a	b	d					